

JONATHAN MA

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Education

Cornell University, College of Engineering, Ithaca, NY

M.Eng. in Systems Engineering

Expected Dec 2026

B.S. in Computer Science, Minor in Robotics (Magna Cum Laude, Dean's Lists)

May 2026

- Coursework: Systems Programming, OS, Intro to Robotics, Computer Vision, Computer Graphics, AI/ML
- TA for Computer Systems (C, RISC V Assembly), OCaml Programming, and Intro to Robotics (ROS) classes
- Honors: Cornell MEng Committee Fellow, Universities Space Research Association Distinguished Undergrad

Experience

NVIDIA: Embedded Software Engineering Intern

May - Aug 2026

- Developed C++ firmware for OpenBMC datacenter out-of-band baseboard management platform with Yocto
- Migrated 800+ multi-platform hardware acceptance tests to a single YAML-based source of truth, reducing manual errors, enabling automated CI/CD validation, and streamlining platform onboarding
- Designed a modular and extensible QEMU/Docker emulation platform for hardware-independent firmware validation with 50+ error injection test scenarios across the network stack

SpaceX: Starshield Flight Software Engineering Intern

May - Aug 2025

- Architected C++ flight and ground software solution spanning 5 satellite systems to deconflict satellite ops
- Conducted over 20 rigorous hardware-in-the-loop tests to ensure mission readiness with embedded Linux
- Organized design reviews and collaborated with 10+ stakeholders across teams, converting ambiguous guidelines to actionable tasks in order to develop a robust solution and testing plan
- Led design reviews with 10+ stakeholders across software, hardware, RF, and payload teams, translating ambiguous mission requirements into actionable tasks and a robust solution and testing plan
- Overhauled satellite integration and testing Python script to increase modularity and testability, taking ownership to drastically reduce risk for critical path test campaign while saving 1000s of engineering hours

Cornell Space Systems Design Studio: Alpha CubeSat Flight Software Lead

Oct 2022 - May 2026

- Promoted to lead a 5-person team after 2 yrs of exemplary performance and initiative to learn/solve problems
- Engineered real-time embedded C++ flight software for [1U CubeSat](#) & [light sail payload](#) on 6-mo LEO mission
- Developed code for memory constrained microcontroller with 32KB flash to interface with avionics firmware/drivers and control GPS, IMU, and LoRa radio peripherals via SPI, I2C, and UART protocols
- Built and deployed [full-stack ground station](#) on Linux system with intuitive React and JavaScript control UI, Python Flask backend API, and Nginx reverse proxy to handle 100+ command & telemetry data points

RTX Collins Aerospace: Systems Engineering Intern, SEPP

May - Aug 2024

- Conducted computer networking research using Information-Centric Networking across a Raspberry Pi fleet
- Extended a 50K-line open-source C++ library by supporting network persistency to improve efficiency

Johns Hopkins University Applied Physics Lab: Ground Software Engineering Intern

May - Aug 2023

- Led team of interns through the design and creation of Angular and Java-based app for parsing binary spacecraft command and telemetry packets for NASA's IMAP and Dragonfly missions, used by 50+ employees
- Applied Agile methodologies and incorporated feedback from project leads throughout development lifecycle

Programming Projects

- Mollusca (C++): Developing swarm formation control algorithm simulator with AprilTag and GTSAM factor graph optimization for autonomous microplastic collection robot
- Birdsong Synthesizer (C): Applied DSP direct digital synthesis algorithm to synthesize birdsongs on ARM-based Raspberry Pi 2040 microcontroller with DAC and ADC using the SPI protocol and GPIO pins
- [MirrorBot](#) (C, Python): Developed mirror control algorithm for robot using ROS, Arduino & Intel RealSense depth camera to detect faces and dynamically adjust reflections to encourage interactions in public spaces
- [Rubik's Cube Solving Robot](#) (C, Python): Arduino powered robot optimized to solve Rubik's Cube in 3-4 sec
 - Created OpenCV pipeline to scan cube and create Rubik's cube mosaic, breadboarded Arduino circuit

Technical Skills

- Languages (Proficient): Python, C++, C, TypeScript, JavaScript, HTML/CSS, Java, OCaml, SQL, Verilog
- Frameworks: Arduino, ROS, OpenCV, Protothreads, NumPy, PyTorch, Pandas, Flask, React, Angular, Firebase
- Tools: Git, Docker, Linux, Bash, PlatformIO, TCP/IP, Elasticsearch, Kibana, Jira, CI/CD, Claude Code, Cursor